

Vinay Kumar

Senior AI Engineer & Agentic Systems Architect

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Professional Summary

Senior AI Engineer and Architect with **5.5+ years of experience** across **Data Science, Predictive Analytics, and Enterprise Agentic AI Platforms**. Proven track record designing foundational AI infrastructure (**gVisor secure sandboxes, event-log execution harnesses, and Model Context Protocol/MCP platforms**) and scaling production products across **Process Automation** (driving a \$300M Telecom transformation deal with a 16-agent HITL pipeline) and **User-Centric Harnesses** exposed via **Web, CLI, VS Code Extension, and SDKs**.

Professional Experience

Infosys — Senior Data Scientist / AI Engineer

Dec 2024 – Present

- Foundational Sandbox Platform:** Architected a DeepAgent-compatible execution sandbox leveraging **gVisor**, enforcing microVM-level isolation, deterministic reproducibility, and strict security governance for untrusted agent code, MCP servers, and dynamic skills.
- Event-Driven Execution Harness:** Engineered a core agent execution runtime featuring **event-log-driven state reconstruction, dynamic context projection, checkpointing, and long-running workflow continuity** for complete observability and auditability.
- MCP Platform & Skill Engine:** Built an AI-assisted authoring, registration, validation, and lifecycle onboarding portal for **Model Context Protocol (MCP) servers** and reusable enterprise skills, standardizing multi-tool connectivity across ecosystems.
- User-Centric Harness Multi-Surface Product:** Designed and shipped a modular agent platform exposed across four primary interaction surfaces: **Web Portal, Terminal CLI, VS Code Extension, and Client SDKs** for workspace-centric agent execution and user onboarding.
- Process Automation Harness (\$300M Deal):** Architected full-stack enterprise agentic workflows for a Tier-1 global telecom operator, boosting process efficiency by **40%** and directly driving the acquisition of a **\$300M AI transformation deal**.
- 16-Agent Production Pipeline:** Engineered an autonomous **16-agent pipeline** with 5 Human-in-the-Loop (HITL) checkpoints, delivering **95% straight-through automation** across complex enterprise operations.
- Proprietary LLM Compiler:** Built an LLM execution optimizer (dynamic prompt planning, semantic caching, and tool routing) that slashed median query latency by **35%** while preserving output accuracy.
- Contract Intelligence System:** Deployed automated contract comparison and clause-risk analysis agents across **5 enterprise clients**, reducing manual legal review time by **60%**.

Xcaliber Infotech — Data Scientist / AI-ML Engineer

Dec 2022 – Dec 2024, Pune

- Summarization Microservice:** Built a high-throughput text summarization microservice using **LLaMA, BART, spaCy, and PyTorch** to process OEM comments for delivery waivers, integrating with warehouse ERPs to improve inventory turnover.
- Enterprise RAG Chatbot:** Developed an enterprise conversational assistant using **LangChain, Hugging Face Transformers, ChromaDB, and Flask**, cutting internal document lookup times by **40%**.
- Accessory Recommender:** Designed a collaborative affinity recommendation engine dynamically matching regional inventory with customer purchase patterns, boosting weekly accessory bundle sales by **25%**.
- API & Telemetry Layer:** Developed secure REST APIs and real-time operations dashboards for role-based analytics, audit tracking, and LLM feedback ingestion.

Concentrix India — Data Analyst

Oct 2015 – Apr 2017, Gurgaon

- Analytics Pipelines:** Built automated Python and SQL ETL pipelines and executive reporting dashboards for operational analytics.
- Predictive Modeling:** Conducted statistical exploratory analysis, regression, and classification modeling to support business forecasting and workforce planning.

Skills

- Foundations & Agentic Systems:** gVisor Agent Sandboxing, Event-Log State Reconstruction, Model Context Protocol (MCP), Skill Authoring Platforms, Multi-Agent Orchestration (CrewAI, LangGraph, AutoGen), LLM Compilers, Semantic Caching, HITL Governance.
- Exposure Surfaces & Tooling:** VS Code Extensions, CLI Tooling, Client SDKs, Web Portals, RESTful APIs, WebSocket Streaming.
- GenAI & LLMs:** LLaMA 2/3, Mistral, GPT-4/Azure OpenAI, Hugging Face Transformers, BERT, RoBERTa, BART, LangChain, RAG Systems, Vector DBs (ChromaDB, Pinecone, FAISS, Redis), Guardrails & LLM-as-a-judge.
- Machine Learning, NLP & Data Science:** Scikit-Learn, PyTorch, TensorFlow, Keras, spaCy, NLTK, Word2Vec, TF-IDF, NER, Regression, Classification, Clustering, Time-Series Modeling.
- Languages, Cloud & Infrastructure:** Python (NumPy, Pandas, Matplotlib), SQL (PostgreSQL, MySQL), FastAPI, Flask, Docker, Kubernetes, AWS (EC2, S3, SageMaker, Lambda), CI/CD Pipelines, Git/GitHub, Linux/gVisor.

Education

PG-Diploma in Artificial Intelligence (PGDAI) — CDAC, Pune, 2022

B.Tech in Mechanical Engineering — JIET, Jind, 2014